

MiniMines Sales Hub — Product Description Document

Version: Engine v2 (2026-07-16) **Repository:** [codeunderscrap/attpatform](https://codeunderscrap.attpatform.com) **Stack:** Python (FastAPI) · React/TypeScript (Vite) · PostgreSQL / SQLite · Docker

1. What Is MiniMines Sales Hub?

MiniMines Sales Hub is an **AI-powered internal sales intelligence platform** for a chemical and battery materials trading company. It ingests raw EXIM (Export-Import) trade data files, runs a multi-stage scoring pipeline, and surfaces ranked shortlists of:

- **Chemical trading opportunities** — which chemicals in MiniMines' portfolio have the highest Attractiveness-to-Trade (ATT) score, and which new chemicals detected in import data are potential expansion opportunities.
- **Battery material procurement intelligence** — which global suppliers and buyers of battery scrap/materials are Tier A/B/C procurement targets.
- **EPR (Extended Producer Responsibility) cross-links** — which companies are importing materials relevant to their EPR obligations and are therefore high-priority sales contacts.

The platform is used daily by the MiniMines sales team to prioritize outreach, validate trading theses, and track market trends — replacing manual spreadsheet analysis.

2. Core Modules

2.1 Chemical ATT Pipeline (ATT = Attractiveness to Trade)

Analyzes every chemical against 8 dimensions of trading attractiveness and produces a score 0–100.

Inputs: - One or more EXIM trade data `.xlsx` files (shipment rows: date, HSN code, seller, buyer, qty, unit, value, description) - A "base portfolio" `.xlsx` file listing the chemicals MiniMines already trades

Outputs: - ATT score per chemical (0–100, Tier A/B/C) - Ranked table: base-portfolio chemicals + opportunity chemicals - 8-dimension breakdown per chemical - Time trend chart (growing / stable / declining) - Buyer & supplier intel tables - Geopolitical risk log - Regulatory risk flag - India incentive bonuses (RoDTEP, drawback) - Downloadable Excel report with methodology sheets

2.2 Battery Procurement Module

Analyzes battery material (scrap, cells, modules) import trade data to rank global suppliers and buyers.

Inputs: - EXIM .xlsx file(s) for battery-related shipments

Outputs: - Entity list (suppliers / buyers) ranked by procurement attractiveness (Tier A/B/C) - Per-category (Li-ion, Lead-acid, E-waste, etc.) price benchmarks - Consistency, volume, geo-ease, and price-index scores per entity - Downloadable Excel report

2.3 EPR Intelligence Module

Cross-links the ATT chemical database with companies' EPR (Extended Producer Responsibility) target obligations.

Inputs: - Configured EPR targets (material → tonnes obligation) per company - ATT run database (shipment history)

Outputs: - Company priority score based on: EPR gap (target – credits) × material importance weight - Cross-linked shipment history: which companies have imported ATT chemicals from recent EXIM runs - AI-generated company research briefs and deal memo drafts (via LLM integration)

2.4 Leads & CRM Module

Lightweight CRM layer for the sales team.

Features: - Lead pipeline with stages: Prospect → Qualified → Proposal → Won/Lost - Notes and activity log per lead - Link leads to ATT chemicals or EPR-identified companies - API-key-gated access for external integrations

2.5 Settings & Administration

- Configurable dimension weights (volume, price, buyers, suppliers, trend, structure, freedom, barrier)
 - Configurable tier thresholds (Tier A ≥ 70, Tier B ≥ 40)
 - Anchor-band floor/ceiling for log-scale dimensions (admin-editable per dimension)
 - EPR material weights
 - LLM provider configuration (API key, model)
 - PIN-gated access (rate-limited, hashed at rest)
 - Retention policy (default: keep forever; opt-in auto-delete with confirmation)
 - Trend exclusion list (months to exclude from regression, e.g. partial current month)
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3. User Flows

Upload & Score a Chemical Run

1. Go to **Upload** → select EXIM file(s) + (optional) custom base portfolio
2. Platform warns if the exact file was previously uploaded (duplicate detection)
3. Pipeline runs in the background with a live progress bar
4. On completion, navigate to **Dashboard** → select the run → view ranked chemicals

Explore Battery Run

1. Go to **Battery** tab → upload EXIM battery files → pipeline runs
2. View supplier/buyer entity tables, filterable by role and tier
3. Download Excel report for offline sharing

EPR Prioritization

1. Go to **EPR** tab → see companies ranked by Priority Score
2. Click **AI Research & Deals** to generate a company brief via LLM
3. Cross-links show which recent EXIM runs contain that company's shipments

Trader Feedback Loop

1. On the **Rankings** page, traders can vote on a chemical: **Confirm / Challenge / Correct**
2. **Challenge** → -2 pts ATT adjustment on next score
3. **Correct** → ±2.5 pts depending on whether the suggested tier is above/below the current score
4. **Confirm** → 0 pts (displayed but not scored — prevents gaming the system)
5. Feedback persists across run deletions — scores stay calibrated over time

4. Technology Stack

Layer	Technology
Backend API	Python 3.11, FastAPI, Uvicorn
Database	PostgreSQL (production) / SQLite WAL (development)
ORM	SQLAlchemy
Data Processing	openpyxl, pandas-free (custom row parser)
NLP Matching	Custom token-similarity (SequenceMatcher) + optional LLM

Layer	Technology
LLM Integration	OpenAI / configurable provider
Frontend	React 18, TypeScript, Vite
UI State	React hooks (no external state library)
Containerization	Docker + docker-compose
Auth	PIN (rate-limited, bcrypt-style SHA256+salt) + API-key header

5. Deployment

```

ATT_Platform/
├── backend/           # FastAPI app, pipeline, DB models
│   ├── main.py       # API routes, auth, upload handling
│   ├── db.py         # SQLAlchemy models, migrations
│   ├── runner.py     # Background job orchestration
│   ├── epr.py        # EPR cross-link logic
│   ├── leads.py      # CRM routes
│   ├── settings.py   # Settings CRUD, PIN hashing
│   ├── rescore.py    # Batch re-score utility (engine upgrades)
│   └── pipeline/
│       ├── engine.py  # Chemical ATT stages 1-6
│       ├── battery.py # Battery scoring pipeline
│       ├── constants.py # Weights, anchor bands, geo events, regulatory DB
│       └── battery_export.py # Excel report generator
├── frontend/         # React/Vite SPA
│   └── src/pages/     # Dashboard, Upload, Battery, EPR, Rankings, Settings...
├── data/             # SQLite file + uploaded EXIM files
├── Dockerfile
├── start.bat         # Windows one-click launcher
└── requirements.txt

```

Quick Start (Windows)

```

double-click start.bat
# or
venv\Scripts\activate
uvicorn backend.main:app --host 0.0.0.0 --port 8000

```

6. Key Design Decisions

Decision	Rationale
Absolute log-anchor scoring (v2)	Cohort-independent — adding/removing one chemical doesn't shift any other chemical's score. Identical raw values score identically.
Per-row hash deduplication	Same shipment appearing in two overlapping EXIM uploads counts once in cross-run dashboards, not twice.
Feedback survives run deletion	Trader domain knowledge accumulates over time and should not silently disappear when old runs age out.
Retention default = 0 (keep forever)	Auto-deleting a sales team's run history by default is the wrong default. Deletion requires explicit opt-in with a confirmation dialog.
confirm = 0 scoring effect	Prevents a coordinated "confirm" campaign from artificially inflating scores beyond what the data supports.
PIN hashed, rate-limited	A 4–6 digit PIN's tiny keyspace means hashing alone can't stop brute force — rate limiting is the real defense; hashing only protects against a copied DB file revealing the PIN.

7. Upcoming / Planned

- EPR engine redesign (Q1–Q10 in docs/EPR_ENGINE_PROBLEMS.md) — awaiting design sign-off
- NLP sub-categorization of "Battery Scrap (Mixed/Other)" descriptions (R4b extension)
- Full cross-run movers dashboard (biggest ATT score changes between runs)
- Mobile-responsive UI

Document generated 2026-07-16 from Engine v2 implementation. For scoring math, see the companion Backend Logic document.